

Abusive Language Detection in Online Conversations and Dating Platforms

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Abstract

The rapid expansion of online communication platforms has led to a significant increase in abusive and manipulative interactions. Traditional moderation approaches are insufficient to handle large-scale user-generated content, necessitating automated detection systems. However, existing models struggle with context-dependent and multi-turn abusive behaviors, particularly in private environments such as dating platforms. This study proposes an integrated framework combining content-based, structural, and manipulation-aware features to improve detection accuracy.

Keywords: Abusive Language Detection, NLP, BERT, Conversational Analysis, Psychological Manipulation

1 INTRODUCTION

Online communication platforms have transformed human interaction but have also enabled the spread of abusive language, including harassment, hate speech, and manipulation. Detecting such behavior is challenging due to its contextual and multi-turn nature.

While transformer-based models have improved detection, they often fail to capture subtle abusive patterns. This limitation is especially critical in dating platforms, where private conversations increase vulnerability.

This work proposes a multi-dimensional detection framework integrating textual, conversational, and behavioral features.

2 BACKGROUND

2.1 Machine Learning Approaches

Early systems relied on keyword filtering, which lacked robustness. Machine learning models improved detection using linguistic features but struggled with context.

2.2 Transformer Models

Models such as BERT and RoBERTa capture contextual semantics and improve classification accuracy. However, they remain limited in detecting implicit abuse.

2.3 Conversational Analysis

Recent approaches model conversations as interaction structures, capturing behavioral patterns that complement textual features.

2.4 Manipulation Detection

Psychological manipulation detection remains challenging due to its subtle and gradual nature, with current models showing high false-negative rates.

3 RESEARCH OBJECTIVES

1. **Evaluate detection models** Assess content-based, structural, and manipulation-aware approaches.
2. **Compare fusion strategies** Analyze Early, Late, and Hybrid fusion techniques.
3. **Improve domain adaptation** Enhance performance on real-world conversational data.
4. **Identify key features** Use feature ablation to determine critical indicators.
5. **Assess deployment feasibility** Evaluate computational efficiency for real-time use.